

FULL NETWORK CONFIGURATION WITH EXPLANATION

1. ROUTER CONFIGURATION (EDGE DEVICE)

Purpose:

- Connect network to ISP (Internet)
- Forward all internal traffic to firewall
- No NAT or VLANs

```
enable
configure terminal
hostname STADIUM-RTR
```

Set router identity

```
interface g0/0/1
ip address 200.10.10.2 255.255.255.252
no shutdown
```

Connect to ISP (WAN link)

```
interface g0/0/0
ip address 172.16.0.1 255.255.255.252
no shutdown
```

Connect router to firewall

```
ip route 0.0.0.0 0.0.0.0 200.10.10.1
```

Default route → sends internet traffic to ISP

```
ip route 10.0.0.0 255.255.0.0 172.16.0.2
ip route 192.168.100.0 255.255.252.0 172.16.0.2
```

Send all internal traffic to firewall

2. ASA FIREWALL CONFIGURATION

Purpose:

- Provide **security + NAT**
- Control traffic entering/leaving network

```
enable
configure terminal
hostname STADIUM-ASA
```

Identify firewall

```
interface g1/1
nameif outside
security-level 0
ip address 172.16.0.2 255.255.255.252
no shutdown
```

Outside interface (untrusted network)

```
interface g1/2
nameif inside
security-level 100
ip address 10.0.0.1 255.255.255.0
no shutdown
```

Inside interface (trusted network)

```
route outside 0.0.0.0 0.0.0.0 172.16.0.1
```

Default route to router

```
object network INSIDE-NET
subnet 10.0.0.0 255.255.0.0
nat (inside,outside) dynamic interface
```

NAT → allows all internal users to access internet

```
access-list BLOCK_PUBLIC extended deny ip 192.168.100.0 255.255.252.0 10.0.0.0
255.255.0.0
access-list BLOCK_PUBLIC extended permit ip any any
access-group BLOCK_PUBLIC in interface inside
```

Security rule:

- Public WiFi → Internal network blocked
 - Everything else allowed
-

3. CORE SWITCH (LAYER 3)

Purpose:

- Perform **inter-VLAN routing**
- Act as **default gateway**
- Central network control

ip routing

Enables Layer 3 functionality

vlan 10,20,30,40,50,60,70,100

Create VLANs (network segmentation)

SVI Interfaces (Gateways)

Example:

```
interface vlan 10
ip address 10.0.10.1 255.255.255.0
no shutdown
```

This becomes:

- Default gateway for VLAN 10

(Same logic applies to all VLANs)

```
interface fa0/1
no switchport
ip address 10.0.0.2 255.255.255.0
no shutdown
```

Routed link to firewall (NOT trunk)

```
ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

Send internet traffic to ASA

```
interface range fa0/2 - 4
switchport mode trunk
no shutdown
```

Connect to distribution switches (carry VLANs)

4. DISTRIBUTION SWITCHES

Purpose:

- Transport VLANs between core and access layer
- No routing, no end devices

Example (DIST-SW1)

```
interface fa0/1  
switchport mode trunk
```

Trunk to core

```
interface fa0/2  
switchport mode trunk
```

Trunk to access switch

```
switchport trunk allowed vlan 10,20,70
```

Restrict VLANs (security + efficiency)

5. ACCESS SWITCHES

Purpose:

- Connect end devices
- Assign VLANs

Example (ACC-SW1)

```
interface fa0/1  
switchport mode trunk
```

Uplink to distribution switch

```
interface fa0/2 - 5  
switchport mode access  
switchport access vlan 10
```

IT devices assigned to VLAN 10

interface fa0/6 - 10
switchport access vlan 20

Staff devices

interface fa0/11 - 15
switchport access vlan 70

CCTV devices

6. DHCP CONFIGURATION (CORE)

Purpose:

- Automatically assign IP addresses

```
ip dhcp pool VLAN20
network 10.0.20.0 255.255.254.0
default-router 10.0.20.1
```

Devices in VLAN 20 get:

- IP address
- Gateway automatically

```
ip dhcp excluded-address 10.0.20.1
```

Prevent gateway IP from being assigned

7. SECURITY SUMMARY

- VLAN segmentation → isolates departments
- Firewall (ASA):
 - NAT for internet
 - Blocks public WiFi access to internal network
- Trunk restrictions → reduce attack surface

8. TESTING OBJECTIVES

- Device gets IP (DHCP)
- Ping gateway
- Inter-VLAN communication
- Internet access
- Public WiFi isolation (blocked)